

MERRYLAND HIGH SCHOOL, ENTEBBE

S.4 MATHEMATICS HOLIDAY WORK

ITEM ONE

Arthur is a graphic designer working on a project that involves transforming shapes digitally. It is known that a certain quadrilateral PQRS was transformed by a transformation matrix $M = \begin{bmatrix} 3 & 2 \\ -2 & 1 \end{bmatrix}$ to form a new quadrilateral P'Q'R'S' with coordinates P'(5,-1) Q'(0,7) R'(-5,-6) and S'(8,-3). Arthur needs to find the coordinates of the quadrilateral PQRS.

Arthur's supervisor, Julius also wishes to transform a certain figure using multiple transformations. The figure first undergoes a scaling transformation of an enlargement with scale factor 2 about the origin and then followed by a rotation about the origin through 90° clockwise. Julius needs to find the matrices of transformation and use them to find the single matrix for the successive transformations.

Task;

- a) Help Arthur determine the coordinates of the quadrilateral PQRS
- b) Help Julius to:
 - (i) State the matrix for the enlargement
 - (ii) State the matrix for the rotation
 - (iii) Determine the single matrix for the successive transformations he can use during his work.

Obtain the single matrix of transformation he can use to transform the final image back onto the object

ITEM 2

Your friend owns a rectangular piece of land of which he wants to construct a house with a rectangular cross section leaving a path 2m around the house base. The path occupies an area of 76 m² and he also wants the dimensions of his house floor to differ by 3m.

He also owns a soft drink depot where he sells Crates of soda and boxes of juice. Tom bought 4 crates of soda and three boxes of juice and paid Shs 110,700. Tom's brother bought 3 crates of soda and 4 boxes of juice and paid Shs 122,400 but he is not contented about the money he paid claiming that he bought almost similar items as Tom but he has paid much money.

Task:

- a) (i) Determine the size of the floor base of your friend's house.
- (ii) How big is your friend's piece of land?
- b) (i) Help Tom's brother to be contented with the payment he made.
- (ii) How much would Ann pay if she buys 2 crates of soda and 2 boxes of water from the same depot?

ITEM THREE

Your mathematics teacher has introduced a new rewarding system based on students' performance in the mathematics assessments. Your teacher has asked you to analyze the scores (in %) of the S2 students which were obtained in the recent mathematics assessment and recorded as given below in order to identify and submit the number of students who qualify for various awards.

28 94 35 78 70 56 57

58 76 60 77 62 84 66

67 61 98 70 96 64 73

74 41 75 62 54 80 83

88 69 90 47 64 70 75

Basing on the new rewarding system, students who score above 85% will receive shs 20,000 each, those who score between 80% and 85% inclusive will receive UGX 10,000 as a contribution towards their pocket.

Those who score below 40% will receive written warnings. Additionally, the school administration has also decided to award those subject teachers whose subject average score is above 70% and if above 50% of your students in the class have scores below 60, you will be requested to be having mandatory weekly tests in order to better the grades.

Task.

- a) Will your mathematics teacher qualify for an award or not? Justify your answer.
- b) How many students qualify for each category of awards?
- c) Will your mathematics teacher be requested to have mandatory weekly tests or not? Justify your answer.

ITEM 4

Your friend who owns a sugar factory in Busoga region wants to buy two types of used motor cycles namely Bajaj and Honda for his sugarcane plantations' supervisors to easy the process of supervising the different sugarcane plantations. A Bajaj costs shs. 2,000,000 and requires shs. 16,000 per month to maintain it. A Honda costs shs. 2,400,000 and requires shs. 10,000 per month to maintain it. The factory owner has shs.18,000,000 to spend on the purchase of the motorcycles and shs. 120,000 per month for maintenance. He would like to buy at least 4 Bajaj motorcycles and at least 2 Honda motorcycles. However, your friend does not know the number of motor cycles of each type that he can buy in order to minimize the maintenance costs.

Task

(a) (i) write down all the mathematical constraints representing the above information.

(ii) Show the constraints on the feasible region.

(b) Determine the possible number of motorcycles of each type that the factory owner should buy in order to achieve his target and hence find the amount saved

ITEM FIVE

In the first week she bought 250 kg of sugar, 180 kg of maize flour, and 140 kg of salt from Lugala Market.

She then bought 100 kg of sugar, and 260 kg of salt from Busiro Market. In the second week she bought 150 kg of sugar and 200 kg of maize flour from Lugala Market and then bought 180 kg of sugar, and 150 kg of salt from Busiro Market. The price per kg of sugar, maize flour and salt in Lugala Market was Shs. 4500, Shs. 2200 and Shs 2000 respectively and the price of sugar, maize flour and salt in Busiro Market was Shs.5500, Shs. 2000 and Shs 1800 respectively.

From, the market, your aunt returned with 12 mangoes in a bag. 5 mangoes are yellow and the rest are green.

Your sister was told to pick two mangoes at random, one at a time, without replacement.

Task:

Help you Aunt;

a) Organize the goods purchased by your guardian in each week.

b) Determine the total amount spent in buying all the items for the two weeks in the two markets?

c) Determine the chance she has to pick;

i) At least one mango that is yellow

ii) Exactly one mango is yellow and the other is green